

PROFESSIONAL WRITING

Chapter 6

Introduction

- There are also some advantages to professional writing, besides those mentioned above.
 - Writing is often used to evaluate an employee for promotion in many jobs.
 - In academia, there is always the concept of publish or perish.
 - Pharmacy technicians are also encouraged to write as a means of advancement.
 - Additionally, writing gives the authors the opportunity to share their knowledge or ideas, obtain gratification or satisfaction, and improve their knowledge.
 - It may even lead to some fame or notoriety in a field

Introduction

- **Challenges to professional writing.**
 - Any significant amount of writing often involves **a lot of potentially frustrating work**, because few people are natural writers.
 - If the author is not proficient at writing, it exposes a person to criticism and possible rejection.
 - Authors may pay to have their article published.
 - At best, the direct financial rewards are likely to be few, unless a best-selling novel is produced.
 - Indirectly, writing may lead to pay increases and promotions.

Steps in Writing

1. Prepare to write

- The first step in writing is to know the **purpose**- why something needs to be written in the first place.
- It is necessary at this time to have a good idea of the **expected endpoint**, which is a good idea, no matter what is being done.
- Topic should be of interest and/or importance to the prospective readers.

Steps in Writing

2. Decide whether there needs to be a coauthor

- Depending on who is working on the project
- An inexperienced **writer** (e.g., a student or someone early in his or her career) would benefit from working with an experienced author.
- Additionally, working with someone will give a different perspective (especially if from many disciplines) and, hopefully, lessen the work for each person.
- Finally, it is sometimes a necessity to include coauthors for political reasons (inappropriate).

Steps in Writing

- **Criteria for an individual to be given credit as an author:**
 - Conception and design of the study, or analysis and interpretation of the data in the study.
 - Writing or revising the article.
 - Final approval of the version that is published.
- Things that do not qualify a person to be listed as an author include:
 - Acquisition of funding
 - General supervision of the research group
- Individuals that do not meet the first qualifications should be listed in the acknowledgments section.

Steps in Writing

3. Know the audience

- it is necessary to know the type of person who will be reading the final document and where it will be published
 - Aim both the writing style and depth of information toward the audience

Types of Technical Writing:

- **Pure technical style:** used by professionals addressing other professionals in the same field.
 - author can use technical jargon, written in third person
 - Great majority of writing done by health professional will be in this style.
 - E.g. articles published on AHSP, NEJM, JAMA, EPJ

Steps in Writing

- **Middle technical style:** used by professionals addressing professionals in other fields.
 - E.g., **report** regarding pharmacy service to PTC,
- **Popular technical style:** used by professionals addressing lay people.
 - Common language is used throughout.
 - E.g., a patient information sheet or patient education materials

Steps in Writing

4. Know the requirements of the publisher

- Specific guidelines are followed by a number of professional journals
- In the case of a journal, directions on the format to follow will be published at least once a year, usually in the first journal publication of the year or on the journal's Web site.

Checklist in Preparation of Written Materials

- Do research first.
- Put yourself in the reader's position.
- Use proper grammar and spelling.
- Make the document look "professional."
- Keep things simple & direct: **KISS** (Keep It Simple, Stupid) principle
- Keep the document short.
- Avoid abbreviations and acronyms.
- Avoid the first person (e.g., I, we, and us).

Checklist in Preparation....

- Use active sentences.
- Avoid slash construction (e.g. he/she use he or she).
- Avoid contractions.
- Avoid sexism
- Cite other references wherever appropriate (and get permission to do so where appropriate).
- Cover things in whatever order is easiest.
- Get everything down on paper before revising.
- Edit, Edit, Edit!

Specific Document Sections

- A typical document consists of three main parts—the *introduction, body, and conclusion*.
- In the case of a clinical study, follow the "**IMRAD**" structure: **Introduction, Methods, Results, and Discussion**.
- Other parts, such as references, tables, figures, and abstracts may also be necessary

Introduction

- *The two most important paragraphs in any document are **the first** and **last**.*
- **Start out strong, to encourage the reader to continue reading.**
- First paragraph should also inform the reader of what they can expect in the rest of the document.
- Introduction should have a clear objective for the existence of the document.
- The introduction should generally not be a conclusion

Body

- Body of the document contains all of the details.
- In a research article, the body may be divided into the **methods, results, discussion, and conclusion**.
- Be concise, yet present all necessary information (do not stray from the subject unless it is absolutely necessary).
- Provide a balanced coverage of the material and avoid unsupported bias.

Body

- Cover the information in a logical order, so that it flows easily from one point to another.
- *If the information is taken from a particular source, even if it is reworded, the **original author** should be given credit via footnotes or endnotes.*
- Avoid extensive **quotations** (copy and pasting of ideas and working of original author).
- **Plagiarism** can be considered the copying of another's words or ideas, without properly giving credit.

Conclusion

- A conclusion should be placed at the end of the body of the document, except for certain documents (e.g., policy and procedures).
- Conclusion should follow logically from the information presented and should serve to summarize that information.
- Conclusion should also correspond with the objective stated in the introduction.

Other Items

- Other items such as tables, graphs, figures, and so forth
 - To supplement or clarify (not distort or misrepresent) and
 - Not to duplicate material in the text portion of a piece of written work.

Submission of the Document

- Once the document is completed, proofread, and edited, it is ready to be submitted, whether this is to a supervisor or a journal.
- For the journal, the author will need to include a cover letter that serves as an introduction to the document.
- It may be necessary to include transfer of copyright forms, conflict of interest disclosures (including financial), or other items.

Revision

- *In many cases, revision of the document will be necessary.*
- This may be due to a difference in opinion or different perception of need.
- Although an author should never change a document, **minor revisions** are often necessary to improve clarity or make the document more appropriate in some other manner.
- The comments given with the request for revision are likely to be helpful, and they should be taken seriously.

Newsletters and Websites

- **Newsletters** have been considered to be a part of any pharmacy practice, but have probably been encountered most frequently in hospitals as a method for communicating DTC actions and other drug-related topics to the medical, pharmacy, nursing, and other health care provider staffs.
- In many cases, newsletters are now replaced by a **Web site**.
- **Define the audience:** who will be reading the newsletter or accessing the web site?

Newsletters and Websites

- **Define the Goals of the Newsletter or Website:** generally includes informing and educating the reader, and also to report news. With Web sites, it can also be to directly sell products or gather information
- **Identify Constraints:** No matter what kind of newsletter or Web site is produced, there are always going to be constraints that limit what it can contain and how good it will be.
 - One of the first constraints is **time**.
 - Another constraint is the **people who can, or will, be involved**
 - **Financial and equipments**

Newsletters and Websites

- **Newsletter/Website Design**
 - “appearance is more important than substance”
 - <http://www.usability.gov/pdfs/guidelines.html>
 - keep the publication consistent.
 - **Use a *template*:** a file that contains material that appears the same from issue to issue.

Newsletter or Website Topics

- Adverse drug reactions
- Calendar of events
- Clinical "pearls"
- Effects of external events on jobs
- Job-related information
- New information sources
- New legal or regulatory requirements
- New services

Newsletter or Website Topics

- News from other departments
- Organization's stand on issues
- Personnel policies
- Pharmacoeconomics
- Pharmacy and therapeutics committee actions and news (major area to be covered)
- Productivity improvement
- Professional announcements
- Review of drugs/drug classes
- Quality assurance

Newsletter Distribution

- A good distribution system must be developed.
- It is important to make sure the readers get the newsletter.
- Ensure readers get the newsletters on a regular "cycle," so that they know when to anticipate the arrival of the publication.

Web Posting

- Consider making the material available on the Internet.
- Text documents, such as posters, are easily placed on a website.
- Full slide presentations can be placed on a website, using streaming audiovisual.

DRUG INFORMATION CONSULT: CHOOSING A TOPIC

- Your topic will be written in the
 - **form of a question;**
 - **one sentence long;**
 - **appear at the top of your paper;**
 - **must be related to a medication** (prescription, over-the-counter or herbal products);
 - **must include the disease state** for which the drug is being used;
- Example:
 - **Is metformin used for the treatment of female infertility?**

Reference Sources

- You will need to use at a **minimum**
 - **Four sources** from
 - the primary literature and
 - the tertiary literature.
- It is traditional and logical to use:
 - **Tertiary literature** when preparing your **introduction**, and
 - **Primary literature** when writing the body of your reply.

LENGTH AND STYLE

- The finished consult should be
 - **2 pages in length** and
 - typed **double-spaced** in **12-point fonts**
 - in a **Word** document.
- It should contain an
 - introduction that is at least a paragraph in length;
 - the body of the paper (including one paragraph for each study you consulted), and
 - a summary or conclusion paragraph

Length and Style

- Your paper should be
 - written in third person** language only.
 - your conclusion should be **evidence-based** and should **clearly answer the question**.

THE INTRODUCTION

- **Purpose:**
 - Is to explain to your reader why the question you have asked is important;
 - why the answer is not universally already known.
- Your introduction will generally be **1-2** paragraphs in length.
- It should provide information about:
 - the disease state or condition for which the drug is used
 - a description of the pharmacology of a drug, where appropriate.

Introduction

- Start with **introducing the disease or condition first**
 - Make good use of statistics or **numbers** to explain to your reader how prevalent the condition is
 - Remember that you can get information about disease **epidemiology from good review articles**
- Then, discuss the treatments for the condition, focusing your reader on the specific treatment about which you will be providing evidence.

Introduction

- The last sentence in the introduction will be a transition sentence.
 - It should make your reader want to read on to examine the evidence regarding the question.
- **Plan to reference at least one common medicine resource**
 - (1) Diseases-specific reference for the first step (Harrison's, Pharmacotherapy, or the Merck Manual) and
 - (2) Tertiary drug resource in your introduction.

Is clonidine used for the treatment of migraine headache?

Please follow the following format:

- **Paragraph 1:** Introduction to the **diseases** (including the definition of the disease, epidemiology or prevalence). Use 1-2 tertiary references here
- **Paragraph 2:** Introduction to the **drug** (what it is? What is its most common indications)
- **Paragraph 3:** **Current therapy of the disease** and the problem with current therapy
- **Paragraph 4:** **Clinical studies** (at least two) of the drug and the disease state. Use 2 or more primary references here
- **Paragraph 5:** Summary

Formulary Management / Drug Evaluation Monograph

Chapter 7

Drug Formulary

- Contains a **list of drugs** that are available under that formulary system, which reflects the clinical judgment of the medical staff.
- List usually contains information on the *dosage forms, strengths, names* (e.g., *generic, trade, and chemical*), and *ingredients* of combination products.
- Many drug formulary publications contain a great deal more material related to the drugs, including a summary of ***indications, side effects, dosing, use restrictions***, and other ***clinical information***.

Drug Formulary

- Goal of the formulary system is to provide a decision-making process leading to the ***selection of medications.***
- **Formulary medications should be the most efficacious and cost-effective agents with the fewest side effects.**
- Other factors should also be taken into consideration, such as the variety of dosage forms available for the medication, estimated use, convenience, dosing schedule, compliance, abuse potential, physician demand, ease of preparation, and storage requirements.

Drug Formulary

- The establishment and maintenance of a drug formulary requires that drugs or drug classes be **objectively assessed based on scientific information** (e.g., efficacy, safety, uniqueness, cost, and other appropriate items), not anecdotal prescriber experience.
- The most logical decision to benefit the patient and the institution should be recommended to the pharmacy and therapeutics (P&T) committee.

Evaluating Drugs for Formulary Inclusion

- Criteria developed for approving addition of a drug to a formulary need to minimally include the following:
 - Indications for use
 - Effectiveness
 - Risks (e.g., adverse effects, drug interactions, and potential for medication errors)
 - Cost
- Valid formulary decisions should be based on objective evidence

Evaluating Drugs for Formulary Inclusion

- The drug evaluation monograph provides a structured method to review the major features of a drug product.
- Once a monograph is prepared, it can easily be used as a structured template or overview of a drug product.
- That allows for easy comparison or contrast to other products that may be used for the same indication or that are in the same product class.
- Commercially prepared monographs can also be obtained from several sources that can be used as is or with modifications to suit the needs of the institution

Evaluating Drugs for Formulary Inclusion

- The quality of the commercial monographs may vary, even from the same publisher, and they may **need extensive updating**.
- Often, writing a new drug evaluation monograph may be easier than improving a commercial monograph.
- When PTC desires to review an entire class of drugs, the drug category review is often another method used.
- **Drug class reviews** are often more lengthy than a single product drug evaluation monograph; however, they can also use a similar structure and format.
- Hospitals, health systems, and managed care organizations review an entire class of drugs on a scheduled basis, which must be at least annually, **according to accreditation standards**.

Drug Evaluation Monograph Sections

SUMMARY PAGE

- The first page of the monograph is essentially a summary of the most important information concerning the drug, and includes a specific recommendation of the action to be taken on the product.
- Some **DTCs** only review this first sheet;
- The summary and recommendation could be placed at the end of the monograph, but it is probably best to keep it on the front to make it easier to refer to during the meeting.

Drug Evaluation Monograph Sections

- While **the decision to add or delete a drug from the formulary is seldom black or white**, a general guideline may be helpful.
- If the drug is less expensive or the same price as others, and **more efficacious** or **safer** — add it to the formulary.
- If the drug is more expensive **without added benefit**, such as increased safety or effectiveness — do not add to the formulary (or delete it from the formulary if it is already on it).

Drug Evaluation Monograph Sections

- The problem comes when the drug is **more expensive and also has more benefits.**
- In that case, the careful analysis of the literature and weighing of the institution's needs must be carried out.
- This is the gray area that has no right answer, but the most appropriate decision must be found.
- The most logical decision to benefit the patient and the institution should be recommended to the PTC.
- **Finally**, therapeutic interchange must be considered. It may be appropriate that the new drug be used to substitute for a less desirable agent, or vice versa.

Drug Evaluation Monograph Sections

BODY OF THE MONOGRAPH

- Many parts of the body of the monograph are self-explanatory from their names and will not be discussed in detail.
- **First**, the body may not always be reviewed by the DTC and, even if presented, it may be covered only briefly.
- The body needs to be written as a means to compile the information for reference and further information.

Drug Evaluation Monograph Sections

- **Second**, efforts must be made to ensure that the drug in question has been adequately compared to other therapies (whether drug, surgical, radiation, or something else).
- The person preparing a monograph must go through each section and ask,
“Have comparisons been made between this drug and the appropriate alternative therapy?”

Drug Evaluation Monograph Sections

- **Third**, every item should be addressed, even if only to state that information was not available or that it is not applicable (e.g. absorption of IV drugs).
- This follows the rule that “if it was not written down, it was not done,” or in this case was not reviewed.
- **Finally**, the source of the information should be mentioned — any important statement of fact must be referenced, or must be suspected of being inaccurate.

Drug Evaluation Monograph Sections

- The **Bioavailability/Pharmacokinetics section** is similar to what would be found in most publications, but the information may be difficult to find for some new drugs.
- **The Dosage Form section** is a good place to point out the limitations in dosage forms available for some drugs.
- **Adverse Effects/Toxicities section:** concentrate on the serious and/or common adverse effects for both the specific drug and the drug class. Whenever possible, incidence and severity should be included.
 - An incidence comparison table

Drug Evaluation Monograph Sections

- patient safety information to be addressed in all monographs. It has been recommended that a list of possible safety problems be compiled. This may include concerns in such areas as ordering, transcribing, order entry, storage, order verification, compounding, dispensing, administration, and monitoring.
- It may be good to consider **Risk Evaluation and Mitigation Strategies (REMS)** information from the FDA in this section.
- Some drugs may be added to the formulary simply because of improved patient safety, even though that comes at an increased cost

Drug Evaluation Monograph Sections

- The final section discusses the **cost comparison**, where the product being reviewed is compared in price to other similar products. Typically, three or four medications (both trade name and generic products) are compared
- Preferably, a **pharmacoeconomic analysis** should be prepared because the seemingly more expensive agent may turn out to be less expensive, overall, as it decreases the length of hospitalization, degree of monitoring, or number of ADE.
- ***While some think that cost is emphasized too much in formulary decisions, it is still an extremely important item.***